

CMSC426_Assignment4_sp26

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Assignment 5:

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1 Affine Structure from Motion (50 points)

1.1 Overview

This problem continues the interest point detection and tracking problem from HW2. Now, you will recover a 3D pointcloud from the image sequence hotel.seq0.png ... hotel.seq50.png. You are encouraged to use your results from HW2, but in case you were not able to complete it, we have also included pre- computed intermediate results in the supplemental material. Submit your code so that we can reproduce your results.

The outline of the affine structure from motion algorithm:

```
[107]: ## Import necessary libraries here (You can add libraries you want to use here)
import os
import random
import cv2
import numpy as np
from scipy.io import loadmat
import matplotlib.pyplot as plt
from matplotlib import cm
%matplotlib inline
```

1.2 Data

WARNING: Colab deletes all files everytime runtime is disconnected. Make sure to re-download the inputs when it happens.

```
[108]: # Download Data -- run this cell only one time per runtime
!gdown 1AORin_YMmWkExjI99vfLYvU_dy-9gFTT
```

```
!unzip "/content/Part2_data.zip" -d "/content/"  
# Load Matches  
data = loadmat('/content/Part2_data/tracks.mat')
```

Downloading...

From: https://drive.google.com/uc?id=1A0Rin_YMmWkExjI99vfLYvU_dy-9gFTT

To: /content/Part2_data.zip

100% 5.44M/5.44M [00:00<00:00, 22.3MB/s]

Archive: /content/Part2_data.zip

replace /content/Part2_data/images/hotel.seq0.png? [y]es, [n]o, [A]ll, [N]one,
[r]ename: A

inflating: /content/Part2_data/images/hotel.seq0.png
inflating: /content/Part2_data/images/hotel.seq1.png
inflating: /content/Part2_data/images/hotel.seq10.png
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inflating: /content/Part2_data/images/hotel.seq38.png
inflating: /content/Part2_data/images/hotel.seq39.png
inflating: /content/Part2_data/images/hotel.seq4.png
inflating: /content/Part2_data/images/hotel.seq40.png

```

inflating: /content/Part2_data/images/hotel.seq41.png
inflating: /content/Part2_data/images/hotel.seq42.png
inflating: /content/Part2_data/images/hotel.seq43.png
inflating: /content/Part2_data/images/hotel.seq44.png
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inflating: /content/Part2_data/images/hotel.seq46.png
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inflating: /content/Part2_data/images/hotel.seq48.png
inflating: /content/Part2_data/images/hotel.seq49.png
inflating: /content/Part2_data/images/hotel.seq5.png
inflating: /content/Part2_data/images/hotel.seq50.png
inflating: /content/Part2_data/images/hotel.seq6.png
inflating: /content/Part2_data/images/hotel.seq7.png
inflating: /content/Part2_data/images/hotel.seq8.png
inflating: /content/Part2_data/images/hotel.seq9.png
inflating: /content/Part2_data/tracks.mat

```

1.3 Code [30 pts]

```

[109]: from operator import length_hint
def perform_rank3_svd(D):
    """
    Perform SVD on the measurement matrix D and enforce a rank 3 approximation.

    Note: Later steps in the algorithm use W3 to build up the motion and
    structure matrices.

    Parameters
    -----
    D : numpy.ndarray
        The measurement matrix of shape (m, n).

    Returns
    -----
    U3 : numpy.ndarray
        The left singular vectors (first 3 columns) of shape (num_frames*2, 3).
    W3 : numpy.ndarray
        The diagonal matrix containing the square roots of the top three
    singular values, of shape (3, 3).
    V3 : numpy.ndarray
        The first 3 columns of V, shape (num_points, 3).
    """
    U, W, V = np.linalg.svd(D)
    U3 = U[:, :3]
    W3 = np.diag(W[:3])
    V3 = V.T[:, :3]
    return U3, W3, V3

```

```

def build_ortho_constraint_system(A):
    """
    Build the orthographic constraint system from the motion matrix A.

    Parameters
    -----
    A : numpy.ndarray, shape (2 * num_frames, 3)
        The uncorrected motion matrix from the SVD step, where the top half
    ↪ (ai) corresponds
        to x measurements and the bottom half (aj) corresponds to y
    ↪ measurements.

    Returns
    -----
    constraint_matrix : numpy.ndarray, shape (3 * num_frames, 9)
        The left-hand side (lhs) constraint matrix, built using outer products.
    ks (`k` in the slides) : numpy.ndarray, shape (3 * num_frames, 1)
        The right-hand side (rhs) vector for the system. The constraints on the
    ↪ squared norms
        are set to 1 (for the first two equations of each frame) while the
    ↪ cross-product
        constraint is set to 0.
    """
    # Separate the motion matrix A into the parts corresponding to x and y:
    num_frames = A.shape[0]//2
    ai = A[0:num_frames]
    aj = A[num_frames:]

    # There are 3 constraints per frame:
    # ||ai||2 = 1, ||aj||2 = 1, and aiT * aj = 0.

    constraint_matrix = np.zeros((3*num_frames, 9))
    ks = np.zeros((3*num_frames, 1))
    # For each frame, fill in the three constraints to finish left-hand side
    ↪ (lhs) and right-hand side (rhs)
    ## Pseudo codes
    for i, (ai_row, aj_row) in enumerate(zip(ai, aj)):
        idx = 3 * i
        constraint_matrix[idx] = np.outer(ai_row, ai_row).flatten() # ->
    ↪ outer product of ai_row with itself (ai_row x ai_row)
        constraint_matrix[idx + 1] = np.outer(aj_row, aj_row).flatten() # ->
    ↪ outer product of aj_row with itself (aj_row x aj_row)
        constraint_matrix[idx + 2] = np.outer(ai_row, aj_row).flatten() # ->
    ↪ outer product of ai_row and aj_row (ai_row x aj_row)

```

```

ks[idx : idx + 3] = np.transpose(np.array([[1,1,0]])) # -> [1, 1, 0]^T

return constraint_matrix, ks
def affineSFM(x, y):
    '''
    Affine Structure from Motion (SFM) Algorithm.

    This function implements an Affine Structure from Motion approach to recover
    the 3D structure of a scene and estimate the camera motion from its 2D
    ↪projections.
    It assumes an orthographic camera model and employs Singular Value
    ↪Decomposition (SVD)
    to enforce a rank-3 constraint, followed by a correction step that applies
    ↪orthographic
    projection constraints.

    The algorithm proceeds as follows:
    1. Normalize the measurements: Subtract the mean from each column of x and
    ↪y so that
        the data is zero mean across each frame.
    2. Construct the measurement matrix D: Stack the transposed normalized x
    ↪and y coordinates.
        This results in a D of shape (num_frames*2, num_points), where each
    ↪pair of rows corresponds
        to the x and y measurements for a given frame.
    3. Perform SVD and enforce a rank-3 model: Decompose the measurement matrix
    ↪and retain only the
        top three singular values and corresponding vectors to satisfy the
    ↪rank-3 condition.
    4. Apply orthographic constraints: Establish and solve a system of
    ↪equations based on the
        orthographic projection properties of the affine camera, then use a
    ↪Cholesky decomposition
        to refine the solution.
    5. Recover the motion and structure matrices:
        - A_corrected: The refined camera motion (affine projection) matrix of
    ↪shape (num_frames*2, 3)
        - S_corrected: The 3D structure (shape) matrix of the scene of shape
    ↪(3, num_points)

    Parameters
    -----
    x : numpy.ndarray, shape (num_points, num_frames)
        A 2D array of x-coordinate feature measurements across frames.

```

```

y : numpy.ndarray, shape (num_points, num_frames)
    A 2D array of y-coordinate feature measurements across frames.

Returns
-----
A_corrected : numpy.ndarray, shape (num_frames*2, 3)
    The refined affine camera motion matrix that maps the 3D structure to the
    observed 2D coordinates.
S_corrected : numpy.ndarray, shape (3, num_points)
    The reconstructed 3D structure matrix, representing the positions of
    feature points in 3D space.
'''

##### YOUR CODE HERE #####
# Step 1:
# Normalize x and y to zero mean.
# Compute the mean for each column (i.e., each frame).
# After normalization, the shape should still be (num_points, num_frames)
x_norm = x - np.mean(x, axis=0)
y_norm = y - np.mean(y, axis=0)

# Step 2:
# Create measurement matrix D.
# The shape of D should be (num_frames*2, num_points)
D = np.vstack([x_norm.T, y_norm.T])

# Step 3:
# Perform SVD and enforce rank 3
# See https://numpy.org/doc/2.2/reference/generated/numpy.linalg.svd.html for
# details.
U3, W3, V3 = perform_rank3_svd(D)

# Step 4:
# Create motion and shape matrices A, S
#  $A = UW^{(1/2)}$ 
#  $S = W^{(1/2)}V^T$ 
A = U3 @ np.sqrt(W3)
S = np.sqrt(W3) @ V3.T

# Step 5:
# Set up the orthographic constraint system to solve for L.
# Each frame contributes 3 constraints:
#  $\|a_i\|^2 = 1$ ,  $\|a_j\|^2 = 1$ ,  $a_i^T a_j = 0$ 
#
# For more details:
# Each constraint can be written as:
# (outer product of two rows) · vec(L) = k
# e.g., using  $a_i = [a \ b \ c]$ ,  $a_j = [d \ e \ f]$ :

```

```

#         outer(ai, aj) → [ad ae af bd be bf cd ce cf]
#
# Or:
#     lhs @ vec(L) = rhs
#
# Finish `build_ortho_constraint_system`
# Make sure passing the sanity check below
# Hint: use np.outer for outer product.
constraint_matrix, rhs = build_ortho_constraint_system(A)

# Step 6:
# Solve the least squares for L
# See https://numpy.org/doc/2.1/reference/generated/numpy.linalg.lstsq.html
# for more details
L_vec = np.linalg.lstsq(constraint_matrix, rhs)[0]
L = L_vec.reshape((3, 3))

# Step 7:
# Cholesky decomposition. See https://numpy.org/doc/2.2/reference/generated/
# numpy.linalg.cholesky.html
C = np.linalg.cholesky(L)

# Step 8:
# Update A and S with the constraints
# to get A_corrected and S_corrected
A_corrected = A @ C
S_corrected = np.linalg.inv(C) @ S

return A_corrected, S_corrected

```

1.3.1 Sanity check - perform_rank3_svd()

```

[110]: # Sanity check for rank-3 SVD
np.random.seed(0) # For reproducibility

D = np.random.randn(8, 10)

# Perform the rank-3 SVD
U3, W3, V3 = perform_rank3_svd(D)

# Print the shapes to check they are as expected.
print("U3 shape:", U3.shape) # Expected: (8, 3) if D is (8, 10)
print("W3 shape:", W3.shape) # Expected: (3, 3)
print("V3 shape:", V3.shape) # Expected: (10, 3)

print(U3)
print(W3)

```

```
print(V3)
```

```
U3 shape: (8, 3)
W3 shape: (3, 3)
V3 shape: (10, 3)
[[ 0.53246692  0.45719775 -0.15921115]
 [ 0.3115632   0.00164329  0.32281968]
 [ 0.43344853 -0.62446345 -0.58126173]
 [-0.16679875 -0.26983316  0.43462257]
 [-0.4874529   0.34449376 -0.55173433]
 [-0.17789561 -0.25325639  0.11268204]
 [-0.3544282  -0.34054681 -0.15616012]
 [ 0.09397544 -0.17151124  0.04672569]]
[[5.1812973  0.          0.          ]
 [0.          4.76627564  0.          ]
 [0.          0.          4.09283553]]
[[ 0.16000216  0.48858191  0.47244524]
 [ 0.31832224 -0.17030302  0.26373828]
 [ 0.50159274 -0.04793424  0.06489539]
 [ 0.19198533  0.79587716 -0.42622534]
 [ 0.46376952 -0.16087835 -0.33118127]
 [-0.16549708  0.08699357  0.36584415]
 [ 0.3630006   0.06973369  0.42800566]
 [-0.20794436 -0.03024835  0.02322414]
 [ 0.37894031 -0.19559239  0.0010515 ]
 [ 0.15342463 -0.1381398  -0.30676682]]
```

Expected output:

```
D shape: (8, 10)
U3 shape: (8, 3)
W3 shape: (3, 3)
V3 shape: (10, 3)
[[ 0.53246692  0.45719775 -0.15921115]
 [ 0.3115632   0.00164329  0.32281968]
 [ 0.43344853 -0.62446345 -0.58126173]
 [-0.16679875 -0.26983316  0.43462257]
 [-0.4874529   0.34449376 -0.55173433]
 [-0.17789561 -0.25325639  0.11268204]
 [-0.3544282  -0.34054681 -0.15616012]
 [ 0.09397544 -0.17151124  0.04672569]]
[[5.1812973  0.          0.          ]
 [0.          4.76627564  0.          ]
 [0.          0.          4.09283553]]
[[ 0.16000216  0.48858191  0.47244524]
 [ 0.31832224 -0.17030302  0.26373828]
 [ 0.50159274 -0.04793424  0.06489539]
 [ 0.19198533  0.79587716 -0.42622534]
```



```

[ 0.46376952 -0.16087835 -0.33118127]
[-0.16549708  0.08699357  0.36584415]
[ 0.3630006   0.06973369  0.42800566]
[-0.20794436 -0.03024835  0.02322414]
[ 0.37894031 -0.19559239  0.0010515 ]
[ 0.15342463 -0.1381398  -0.30676682]]

```

1.3.2 Sanity check - build_ortho_constraint_system()

```

[111]: # Sanity check for build_ortho_constraint_system()
A_simple = np.array([
    [1, 2, 3], # ai for frame 1
    [4, 5, 6], # ai for frame 2
    [7, 8, 9], # aj for frame 1
    [10, 11, 12] # aj for frame 2
])
constraints_lhs, constraints_rhs = build_ortho_constraint_system(A_simple)

expected_lhs = np.array(
    [[ 1.,  2.,  3.,  2.,  4.,  6.,  3.,  6.,  9.],
     [ 49., 56., 63., 56., 64., 72., 63., 72., 81.],
     [ 7.,  8.,  9., 14., 16., 18., 21., 24., 27.],
     [ 16., 20., 24., 20., 25., 30., 24., 30., 36.],
     [100., 110., 120., 110., 121., 132., 120., 132., 144.],
     [ 40., 44., 48., 50., 55., 60., 60., 66., 72.]]
)
expected_rhs = np.array(
    [[ 1.],
     [ 1.],
     [ 0.],
     [ 1.],
     [ 1.],
     [ 0.]]
)
print("Constraints LHS:")
print(constraints_lhs)
print("Constraints RHS:")
print(constraints_rhs)

assert np.allclose(constraints_lhs, expected_lhs), "Constraints LHS not correct"
assert np.allclose(constraints_rhs, expected_rhs), "Constraints RHS not correct"

print("Passed!")

```

Constraints LHS:

```

[[ 1.  2.  3.  2.  4.  6.  3.  6.  9.]
 [ 49. 56. 63. 56. 64. 72. 63. 72. 81.]
 [ 7.  8.  9. 14. 16. 18. 21. 24. 27.]

```

```

[ 16.  20.  24.  20.  25.  30.  24.  30.  36.]
[100. 110. 120. 110. 121. 132. 120. 132. 144.]
[ 40.  44.  48.  50.  55.  60.  60.  66.  72.]]
Constraints RHS:
[[1.]
 [1.]
 [0.]
 [1.]
 [1.]
 [0.]]
Passed!

```

1.3.3 Data preprocessing

```

[112]: track_x = data['track_x']
       track_y = data['track_y']
       print(track_x.shape)
       print(track_y.shape)

       # Remove the nan value
       valid_indices = ~np.isnan(track_x).any(axis=1) & ~np.isnan(track_y).any(axis=1)
       valid_x = track_x[valid_indices]
       valid_y = track_y[valid_indices]
       print(valid_x.shape)
       print(valid_y.shape)

(500, 51)
(500, 51)
(400, 51)
(400, 51)

```

1.4 Write-up [20 pts]

- Plot the predicted 3D locations of the tracked points for 3 different viewpoints. Choose the viewpoints so that the 3D structure is clearly visible.

```

[113]: print(valid_x.shape)
       A, S = affineSFM(valid_x, valid_y)

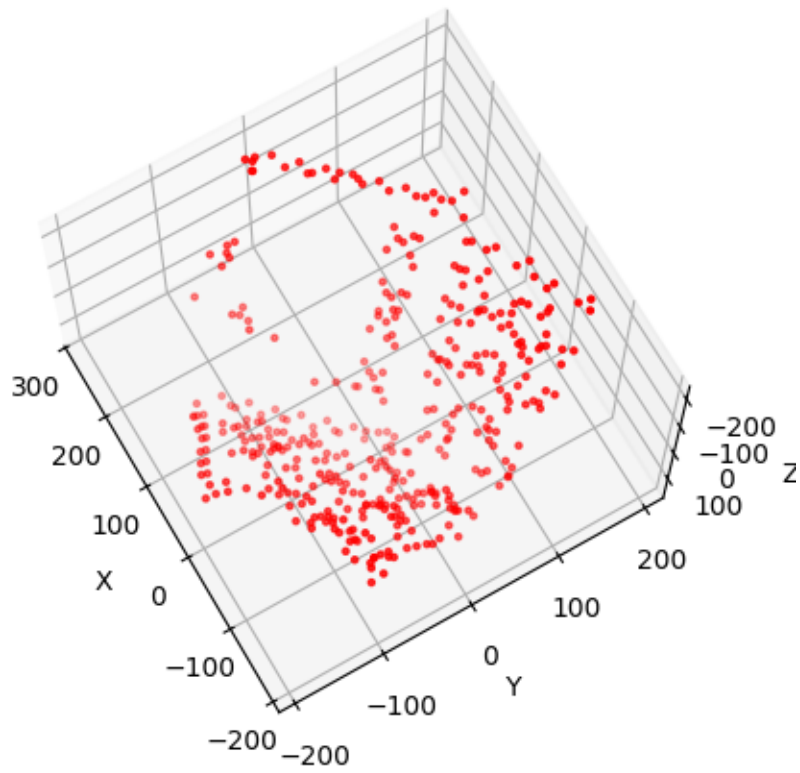
       # Plot the predicted 3D locations of the tracked points for 3 different
       ↪viewpoints.
       # Choose the viewpoints so that the 3D structure is clearly visible.
       viewpoints = [(-110, 30), (90, 180), (-75, 90)]

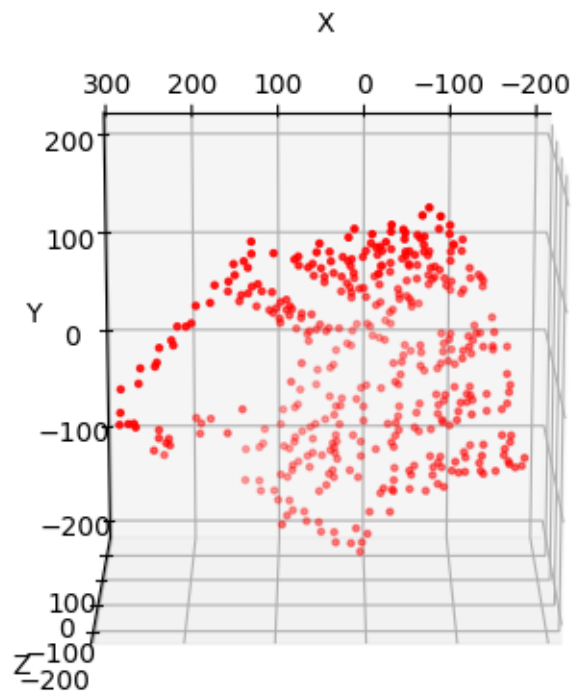
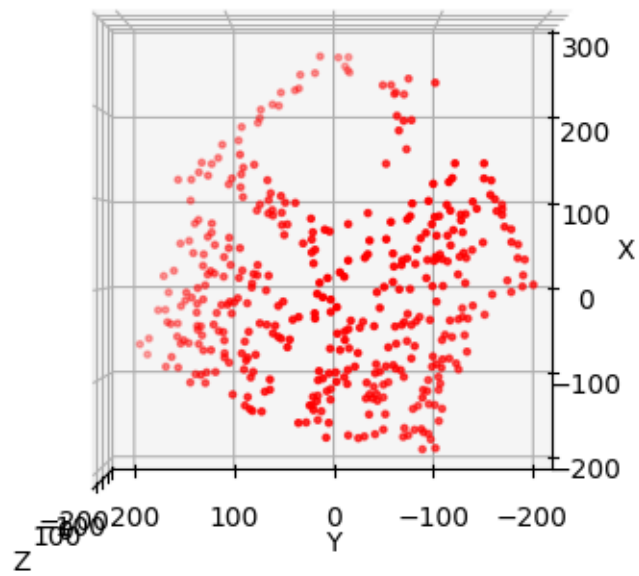
       for elev, azimuth in viewpoints:
           fig = plt.figure()
           ax = fig.add_subplot(111, projection='3d')
           ax.scatter(S[0, :], S[1, :], S[2, :], c='r', marker='.')

```

```
ax.view_init(elev=elev, azim=azim)
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
plt.show()
```

(400, 51)

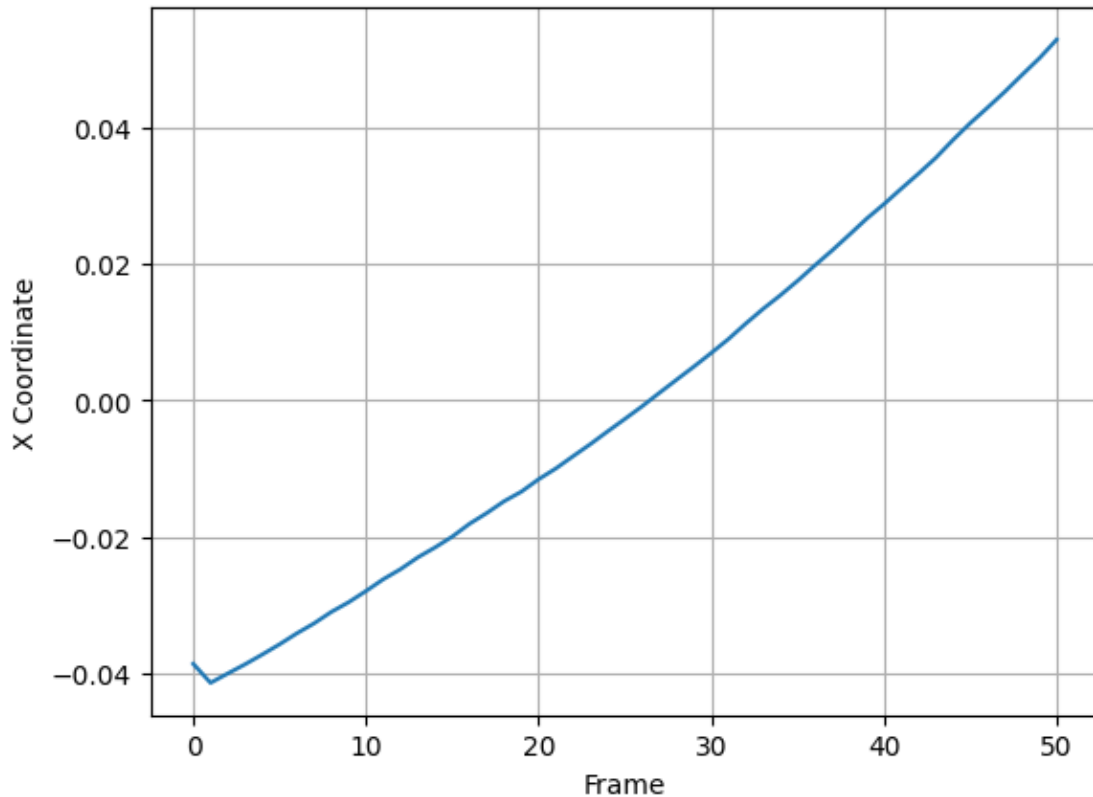


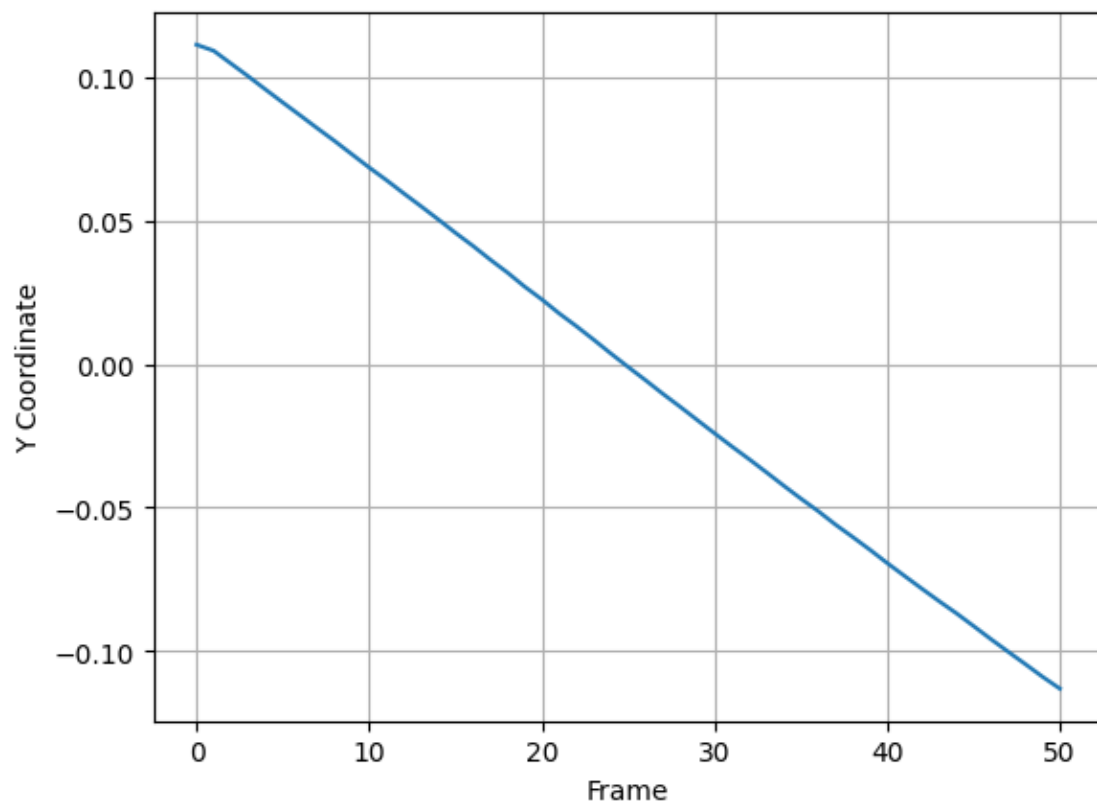


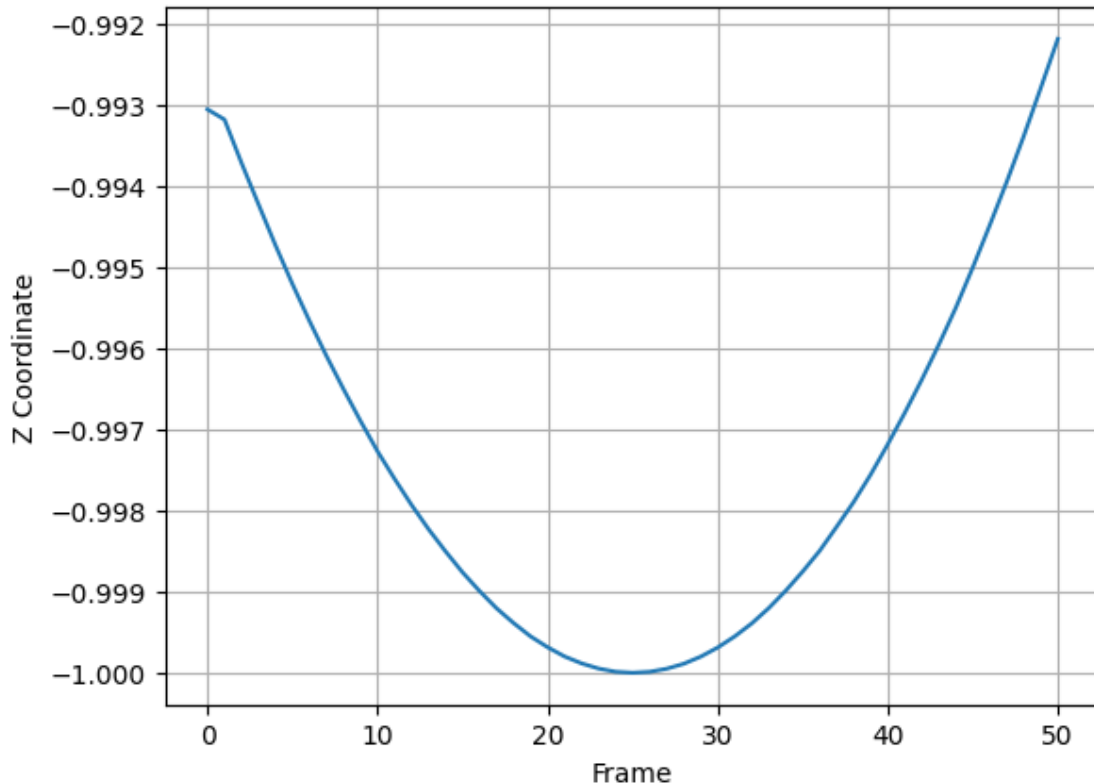
- Plot the predicted 3D path of the cameras. The camera position for each frame is given by the cross product $a_k = a_i \times a_j$. Normalize a_k to be unit length for consistent results. Give 3 plots, one for each dimension of a_k

```
[114]: # Plot the predicted 3D path of the cameras. The camera position for each frame
# is given by the cross product  $a_k = a_i \times a_j$ .
# Normalize  $a_k$  to be unit length for consistent results. Give 3 plots, one for
# each dimension of  $a_k$ 
ai = A[:A.shape[0]//2, :]
aj = A[A.shape[0]//2:, :]
ak = np.cross(ai, aj)
ak /= np.linalg.norm(ak, axis=1)[:, np.newaxis]

for i, label in enumerate(['X', 'Y', 'Z']):
    plt.plot(ak[:, i])
    plt.xlabel('Frame')
    plt.ylabel(f'{label} Coordinate')
    plt.grid(True)
    plt.show()
```







1.4.1 Hint

- Do not use existing structure from motion code, such as found in OpenCV.
- The provided file `tracks.mat` contains the tracked keypoints over 51 frames.viewpoints.
- Reference:
 - Tomasi and Kanade. Shape and Motion from Image Streams under Orthography: a Factorization Method. 1992

2 Save as a PDF file

Make sure you have run Dependencies below. It will take a while. If the following code does not work for you, go `File > Print > Save`.

```
[ ]: !apt-get update
!dpkg --configure -a
!apt-get install -y texlive texlive-xetex texlive-latex-extra pandoc
!pip install pypandoc
```

```
0% [Working]          Get:1 https://cli.github.com/packages stable
InRelease [3,917 B]
0% [Waiting for headers] [Connecting to security.ubuntu.com (185.125.190.82)] [
Get:2 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease
```

```

[3,632 B]
Hit:3 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:6 https://cli.github.com/packages stable/main amd64 Packages [354 B]
Get:7 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]
Get:8 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:9 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ Packages [89.0
kB]
Get:10 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
[18.1 kB]
Hit:11 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Get:12 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [2,970 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4,264
kB]
Get:14 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [10.0 MB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages
[1,623 kB]
Get:16 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64
Packages [7,015 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages
[7,232 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages
[35.9 kB]
Get:19 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy/main amd64
Packages [38.8 kB]
Get:20 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages
[3,929 kB]
Get:21 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages
[1,311 kB]
Fetched 39.0 MB in 5s (7,173 kB/s)
Reading package lists... Done
W: Skipping acquire of configured file 'main/source/Sources' as repository
'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide
it (sources.list entry misspelt?)
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!

```



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Setting up libjbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.3) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-latex-base (2021.20220204-1) ...

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Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.12) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.12) ...
Setting up ruby (1:3.0~exp1) ...
Setting up rake (13.0.6-2) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.

```

```

Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
texlive is already the newest version (2021.20220204-1).
texlive-latex-extra is already the newest version (2021.20220204-1).
The following additional packages will be installed:
    texlive-plain-generic tipa
Suggested packages:
    default-jre-headless tipa-doc
The following NEW packages will be installed:
    texlive-xetex tipa
The following packages will be upgraded:
    texlive-plain-generic
1 upgraded, 2 newly installed, 0 to remove and 39 not upgraded.
1 not fully installed or removed.
Need to get 42.8 MB of archives.
After this operation, 87.2 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 42.8 MB in 3s (14.4 MB/s)
(Reading database ... 154123 files and directories currently installed.)
Preparing to unpack .../texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) over (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive-xetex (2021.20220204-1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for tex-common (6.17) ...
Running mktexlsr. This may take some time... done.
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time...

```

You will be prompted for permissions to Google Drive.

```
[116]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

Change the path accordingly below. You will find PDF file on your left hand side.

```
[ ]: # Feel free to change the path
# !cp <path-to-this-notebook-in-GoogleDrive> ./
##### Reminder #####
# Make sure that you can find your ipynb file under /content/drive/MyDrive or /
  ↳ content/drive/MyDrive/Colab_Notebooks or your path/to/ipynb_file
#####
!cp /content/drive/MyDrive/CMSC426/CMSC426_Assignment4_sp26.ipynb ./ # subject_
  ↳ to change
!jupyter nbconvert --to PDF "CMSC426_Assignment4_sp26.ipynb"
# PDF file will be on the left hand side bar
```

```
[117]:
```